

Name: _____

Class: _____

Date: _____

Homework 4.1

The Internet

Score: ____ / 8

Multiple Choice

1. A protocol is...

- A. a password
- B. a type of computer
- C. a web page
- D. an agreed-upon set of rules that specify the behavior of a system

2. A packet contains...

- A. a chunk of data and metadata used for routing
- B. only the sender's name
- C. an entire web page always
- D. only error messages

3. What does a router do?

- A. Stores websites
- B. Forwards packets toward their destination
- C. Encrypts passwords
- D. Prints documents

4. Bandwidth is usually measured in...

- A. bytes per file
- B. bits per second
- C. pixels
- D. hertz of the screen

Written 1

Describe the journey of a message sent across the Internet, using the words packet, IP address, router, and TCP.

Written 2

Compare TCP and UDP and give an app that would use each.

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Homework 4.2

Fault Tolerance

Score: ____ / 8

Multiple Choice

1. Redundancy is...

- A. a type of compression
- B. deleting duplicate files
- C. extra components that can be used if other components fail
- D. a slow network

2. A fault-tolerant system...

- A. never has failures
- B. cannot be repaired
- C. has only one path
- D. keeps working even when some components fail

3. Why does the Internet have multiple paths between devices?

- A. To store more data
- B. To make it slower
- C. To save money
- D. So data can be rerouted if a path fails

4. What is the main trade-off of adding redundancy?

- A. It decreases reliability
- B. It makes the network smaller
- C. It removes all failures
- D. It requires more resources and cost

Written 1

Network: Library-School, School-Gym, Gym-Pool, Library-Gym. Which single connection failure would disconnect the Pool? Suggest one connection to add so no single failure disconnects any building.

Written 2

Explain why redundancy makes the Internet more reliable, and give one cost of redundancy.

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Homework 4.3

Parallel and Distributed Computing

Score: ____ / 8

Multiple Choice

1. Sequential computing performs operations...

- A. one at a time, in order
- B. all at the same time
- C. on many devices
- D. randomly

2. Parallel computing...

- A. breaks a program into smaller operations, some performed simultaneously
- B. runs only one task at a time
- C. requires the Internet
- D. is always slower

3. Distributed computing uses...

- A. one processor
- B. multiple devices to run a program
- C. only phones
- D. no network

4. Speedup is calculated as...

- A. $\text{parallel time} \div \text{sequential time}$
- B. $\text{sequential time} \div \text{parallel time}$
- C. $\text{sequential time} - \text{parallel time}$
- D. $\text{number of processors} \times \text{time}$

Written 1

Tasks: 25 s, 35 s, 15 s, 45 s (independent).
Find the time on one processor, the minimum time on two processors (show the split), and the speedup.

Written 2

Explain why adding more and more processors eventually stops making a program faster.
